05 Ch 05 — Security Valuation

Asked in last 3 papers: CAPM + constant growth (Sep 25, 6m) · bond refunding NPV (Sep 25, 6m) · FCFF/FCFE (Sep 25, 7m) · repo legs MCQs (Jan 26, 6m) · H-model (May 26, 4m) · two-stage FCFE + implied growth (May 26, 5m).

1. Overview of Valuation

→ **Investment** = commitment of funds to earn a return that pays for (i) **time** funds are locked, (ii) expected **inflation**, (iii) **risk**.

→ Estimate **intrinsic value** and compare with market price. Returns may be earnings, cash flows, dividends, interest, interest on interest, capital gains.

→ Same **core principles** for all assets (real or financial); only details and uncertainty differ.

2.1–2.2 Required Rate of Return; Discount Rate

→ **Required return** = minimum return expected over a period = **opportunity cost / cost of capital** (highest return forgone on similar-risk investment). Often used interchangeably with expected return.

→ **Discount rate** = rate to find PV of future cash flows = risk-free rate + risk premium. Each cash flow can have its own rate (inflation, maturity, default) — **term structure**: upward slope = longer maturity, more inflation/liquidity/default risk.

→ Many rates can be replaced by **one equivalent single rate** giving the same PV.

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SM Example (2.2): CF ₹100, 200, 300, 400, 500 at 2.0%, 3.2%, 3.6%, 4.8%, 5.0% → PV **₹1,278.99**; single equivalent rate **4.4861%**.

2.3 Internal Rate of Return

→ **IRR** = rate equating PV of future cash flows to **market price**. = average annual return, **assuming reinvestment at the IRR**.

→ Reinvest below IRR → realised return < IRR. Realised return = (Total cash at end ÷ Investment)$^{1/n}$ − 1 (app-written general form of the SM working).

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SM Example (2.3): −20,000; +3,000 (Y1); +23,000 (Y2) → IRR **15%**. Reinvest 3,000 at 10% → total 26,300 → (26300/20000)$^{1/2}$ − 1 = **14.67%**. Reinvest at 15% → 26,450 → exactly **15%**.

3–3.1 Equity Risk Premium; Explanation

→ **ERP** = excess return of equity over a **risk-free** rate (e.g. tax-free government bonds) — compensation for higher risk; based on **risk-reward trade-off**.

→ Changes with portfolio risk and over time; **theoretical** — only estimated from past stock and bond market data; varies with time frame and method.

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SM Example (3.1): Govt bond 7%, stock 15% → ERP **8%**.

3.2 Calculating the Equity Risk Premium

Formula	Terms / logic
$R_x = R_f + \beta_x(R_m - R_f)$	CAPM. R_x expected return on x; R_f risk-free; β_x systematic risk of x; R_m market return.
Market risk premium = $R_m - R_f$	Premium of the market ($\beta = 1$).
ERP = $R_x - R_f = \beta_x(R_m - R_f)$	Premium of that stock; if x = market, ERP = market premium.

4. Required Return on Equity

Asked: Sep 2025 Q2(a) 6m

→ Investors judge an asset by its contribution to **systematic risk** (not diversifiable). CAPM = relatively **objective** estimate; widely used.

→ Return on share A = R_f + β × Market risk premium.

→ β from data (used in TYK P17): **β = Cov(security, market) ÷ Variance of market**.

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SM Example (4): R_f 5%, β 1.5, MRP 4.5% → 0.05 + 1.5(0.045) = **11.75%**.

5–5.1 Discount Rate Selection; Nominal vs Real Cash Flow

→ Cash flows to **equity** (after debt claims and taxes) → discount at **required return on equity**. Cash flows to **all stakeholders** → **cost of capital**.

→ **Nominal** cash flow: no inflation adjustment. **Real** cash flow: adjusted for inflation. Low inflation/short term → almost same; high inflation → nominal > real.

5.2 Discount rate selection in Equity Valuation

→ Nominal CF ↔ nominal rate; real CF ↔ real rate.

→ Equity valuation uses only **nominal** cash flows and nominal rate — tax on corporate earnings is stated in nominal terms, so it reflects taxes accurately.

→ CF to equity → nominal cost of equity; CF to all capital providers → **nominal after-tax WACC**.

6–6.1 Valuation of Equity Shares; Dividend Based Models

Asked: Sep 2025 Q2(a) 6m

Asked: May 2026 Q1(b) 4m

→ Three approaches: (1) **Dividend** based, (2) **Earning** based, (3) **Cash flow** based.

→ Dividend model assumptions: (a) dividend paid **annually**; (b) first dividend at **end of year 1**; (c) sale at end of a year at **ex-dividend** price.

→ Value = PV of expected dividend stream at K_e (required return / opportunity cost).

Formula	Terms / logic	SM figures
One-year: $P_0 = (D_1 + P_1) \div (1 + K_e)$	Dividend and sale price both received at end of year 1.	D_1 6, P_1 36, K_e 20% → **₹35**
Zero growth: $P_0 = D \div K_e$	Same dividend forever = perpetuity.	—
Constant growth (Gordon): $P_0 = D_1 \div (K_e - g) = D_0(1+g) \div (K_e - g)$	D_0 just paid, D_1 next; g forever; needs $K_e > g$. Rearranged (as in TYK P6 answer): **$K_e = D_1/P_0 + g$**.	TYK P8, P9, P13, P15

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TYK P8: K_e = 8 + 1.5(12 − 8) = **14%**; P_0 = 3(1.10)/(0.14 − 0.10) = 3.30/0.04 = **₹82.50**. **TYK P9:** K_e 9 + 1.5(13 − 9) = 15% → 2.00/0.08 = **₹25.00** (alt. 2.00(1.07)/0.08 = ₹26.75); β 1.75 → K_e 16% → **₹22.22** (alt. ₹23.78). **TYK P13:** k = 9 + 0.75(15 − 9) = 13.5%; P = 2.5 × 1.1/0.035 = **₹78.57**. **TYK P15:** K_e 13.8% → 2.14/0.068 = **₹31.47**; all changes together: 9.18% + 1.3(13% − 9.18%) = 14.146%, g 10% → 2.20/0.04146 = **₹53.06**, alt. 11% + 1.3(15% − 11%) = 16.20% → **₹35.48**; separately: inflation +2% → R_x 15.80% → **₹24.32**; g 10% → **₹57.89**; β 1.3 → 14.2% → **₹29.72**.

Note (app-written): TYK P9 main answer divides D_0 (2.00) by $K_e - g$; the "alternative" with D_1 = 2.00(1.07) follows the SM formula. TYK P15(ii) "all together" gives two ICAI versions (₹53.06 and ₹35.48); the second (R_f and R_m both +2%) matches the separate inflation working (R_x 15.80%).

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TYK P5 (holding period): D_0 14, g 9%: D_1 15.26, D_2 16.63, D_3 18.13, P_3 360, K_e 13% → 13.50 + 13.02 + 12.56 + 249.48 = **₹288.56**; (ii) 9% forever → 15.26/0.04 = **₹381.50**; (iii) P_3 = 18.13 × 1.09/0.04 = 19.76/0.04 = **₹494**. **TYK P14 (bonus 1:5 in yr 4, 5% costs):** dividends 20 × 3 yrs, then 24; sale 900 × 1.2 × 0.95 = 1,026 → PV **₹563.68** vs cost 500 × 1.05 = 525 → gain ₹38.68, buy; max price 563.68 × 100/105 = **₹536.84**.

Variable growth — *many growth rates allowed, but* ***one rate must be for infinity****. Company stages: growth, transition, maturity.*

Formula	Terms / logic
Two-stage: $P_0 = \Sigma_{t=1..n}\, D_0(1+g_1)^t \div (1+K_e)^t + P_n \div (1+K_e)^n$	PV of super-growth dividends + PV of price at end of super growth.
$P_n = D_0(1+g_1)^n(1+g_2) \div (K_e - g_2)$	Gordon at end of year n using D_{n+1}. g_1 finite/super growth; g_2 normal growth; P_n = price at beginning of normal growth. Discount P_n by n years (not n+1).
Three-stage: PV(high-growth dividends) + PV(transition dividends) + PV(price at start of stable growth)	High growth → advantage depleted by competition → gradual decline (transition) → stable growth.
H-Model: $P_0 = D_0(1+g_n) \div (r - g_n) + D_0 \times H \times (g_c - g_n) \div (r - g_n)$	Value at normal growth + **premium for abnormal growth**. g_n long-run normal growth; g_c current (initial short-term) growth; **H = half of the transition period**; growth declines **linearly** over 2H. Also usable for FCFE.

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TYK P3: D_0 4; D_1 4.80, D_2 5.76, D_3 6.336; TV = 6.336/(0.15 − 0.10) = 126.72 at end of yr 2 → 4.80 × 0.8696 + 5.76 × 0.7561 + 126.72 × 0.7561 = **₹104.34**. **TYK P17:** β = 30%/24% = 1.25; K_e = 11 + 1.25(18 − 11) = **19.75%**; dividends 2.80, 3.92, 5.49, 7.68, 10.76 → PV 16.36; TV PV = 10.76(1.15)/(0.1975 − 0.15) × 0.406 = 105.77 → **₹122.13**. **TYK P6 (K_e from price):** (i) 1.12/20 + 12% = **17.6%**; (ii) 20% for 5 yrs then 10%: at 18% PV ₹20.23, at 19% ₹17.89 → 18% + (0.23/2.34) × 1% = **18.10%**. **TYK P16:** now 2(1.06)/0.02 = **₹106**; with project (no dividend 3 yrs, D_4 2.50, g 7%) 2.50/0.01 × 1/$(1.08)^3$ = **₹198.46**; prices ₹214.33 (yr 1), ₹231.48 (yr 2), ₹250 (yr 3); sell 10, 9, 8 shares → ₹2,143.30, ₹2,083.32, ₹2,000; 973 shares left → yr 4 income **₹2,432.50**.

Note (app-written): TYK P16 gives ₹198.46 and then ₹198.45 for the same figure (rounding).

6.2 Earning Based Models

→ Investor may forgo cash dividend for higher earnings on retained earnings → higher future dividend growth → value on **earnings**.

→ **Gordon's assumptions:** (i) return on retained earnings constant; (ii) retention ratio constant.

Formula	Terms / logic	SM figures
Gordon: Value = $EPS_1(1 - b) \div (K_e - br)$	b retention ratio; r return on equity; $EPS_1(1 - b)$ = dividend; growth = b × r (TYK P1: g = 0.6 × 0.15).	TYK P1
Walter: Value = $[D + (E - D)\, r/K_e] \div K_e$ (TYK P1 answer prints it as $V_c = [D + (R_a/R_c)(E - D)] \div R_c$)	D dividend; E EPS; r (= R_a) return on retained earnings; K_e (= R_c) capitalisation rate. Retained (E − D) earns r, capitalised at K_e.	TYK P1
PE multiplier: Value = EPS × PE ratio	PE from similar company/industry, adjusted for company features.	TYK P4, P11
EPS = (PAT − Preference dividend) ÷ No. of equity shares	EPS is for **equity** only.	TYK P18

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TYK P1: EPS = 137.8 × 0.15 = 20.67; D = 0.40 × 20.67 = 8.27; g = 0.6 × 0.15 = 0.09 → Gordon 8.27/(0.18 − 0.09) = **₹91.89**; Walter [8.27 + (0.15/0.18)(20.67 − 8.27)]/0.18 = 18.60/0.18 = **₹103.35**. **TYK P4:** (i) 2.50(1.02)/(0.105 − 0.02) = **₹30**; (ii) PE = 1/ROE = 1/9% = 11.11 → 2.25 × 11.11 = **₹25.00**; earnings growth 2.25 × 1.02/0.07 = **₹32.79**; both < ₹35 → **overvalued**. **TYK P11:** EPS = 5/0.45 = 11.11, D 6.11, price 11.11 × 8 = 88.88 → K_e = 6.11/88.88 + 15% = **21.87%**; g 16% → 6.11/(21.87 − 16)% = **₹104.08**; K_e 20%, g 19% → **₹611.00**.

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TYK P12 (DDM from ratios): Sales 600 × 1.1 = 660; op. cost 594; EBIT 66; interest 175 × 8% = 14; EBT 52; tax 20.80; EAT **31.20**; dividend 5.20; retained 26.00 (₹ lakh). ROE = 31.2/400 = 7.8%; SGR = 0.078 × 0.8333 = **6.5%** (or 0.078 × 0.8333/(1 − 0.078 × 0.8333) = 6.95%). DPS 0.52 → P_0 = 0.52(1.065)/0.085 = **₹6.51** (or ₹6.91) < CMP ₹14 → overvalued, do not invest.

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TYK P7 (yield-based value, Sun Ltd): PAT 15,80,800; pref div 3,20,000; equity div 6,40,000; retained 6,20,800. Coverage = (15,80,800 + 7,68,000)/(7,68,000 + 3,20,000) = **2.16 times** (or 1.91 with interest net of tax). Gearing = (40 + 64)/(80 + 32) = **0.93**. Yield = 50% × 6,40,000 + 5% × 6,20,800 = 3,51,040 → 4.39%. Expected yield = 9.60 + 0.84 (coverage: 3.00 − 2.16) + 0.36 (gearing 0.18 × 2%) = **10.80%**. Value = 4.39/10.80 × 100 = **₹40.65**.

Note (app-written): SM p. 55 prints the gearing premium as (0.75 − 0.93)(2%) = 0.18 (2%); the difference is 0.93 − 0.75 = 0.18. The 0.36% used is correct.

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TYK P18 (buyback): 1.1P(10 lakh − 27 lakh/P) = 210 lakh → 11P − 29.7 = 210 → P = **₹21.79**; shares bought 27/21.79 = 1.24 lakh (123910); EPS 3 × 10/8.76 = **₹3.43**.

6.3–6.3.1 Cash Flow Based Models; FCFF

Asked: Sep 2025 Q6(a) 7m

→ DDM ignores cash needed for **long-term capex** and **short-term working capital** → not the true free cash flow. Free cash flow models discount cash left after meeting these.

→ **FCFF** discounted at **cost of capital (K_0)** → value of the **firm**. **FCFE** discounted at **K_e** → value of **equity** (like DDM).

FCFF from	Formula
(a) Net income	NI + Interest(1 − t) + Depreciation ∓ Capex ∓ ΔNon-cash NWC
(b) EBIT	EBIT(1 − t) + Depreciation ∓ Capex ∓ ΔNon-cash NWC
(c) EBITDA	EBITDA(1 − t) + Depreciation × t ∓ Capex ∓ ΔNon-cash NWC (Dep × t = tax shield on depreciation)
(d) FCFE	FCFE + Interest(1 − t) + Principal repaid − New debt issued + Preferred dividend (add back what went to/came from non-equity providers)
(e) Cash flows	CFO + Interest(1 − t) ∓ Capex (US GAAP; if interest already adjusted in CFO, do not add again)

→ **Capex** (one year) = Purchase of fixed assets − Sale of fixed assets (investing activities).

→ **ΔNon-cash WC:** (1) WC = CA − CL; (2) Non-cash NWC = CA − Cash & bank − CL; (3) same for previous year; (4) change = current − previous; (5) **increase** = more WC needed → **deduct** from FCFF.

→ Value of firm: **one-stage** = PV of stable-period FCFF; **two-stage** = PV explicit (or transition) period FCFF + PV stable period FCFF; **three-stage** = PV explicit + PV transition + PV stable.

6.3.2 Free Cash Flow to Equity (FCFE)

Asked: Sep 2025 Q6(a) 7m

Asked: May 2026 Q5(a) 5m

→ Cash for equity holders after operating expenses, interest, **net debt** obligations and reinvestment (WC, capex).

→ FCFE = Net income ∓ Capex + Depreciation ∓ ΔNon-cash NWC + New debt issued − Debt repayments + Net issue of preference shares − Preference dividends.

→ Or FCFE = Net profit + Dep − ΔNWC − Capex + New debt − Debt repayment + Net pref issue − Pref dividend (Capex = addition to fixed assets to sustain the base).

→ FCFE can be valued by **multistage growth** (two-stage, three-stage, H-model) exactly like dividends.

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TYK P10: shares 1,300/40 = 32.5 cr; EPS 290/32.5 = **₹8.923**; FCFE = 8.923 − [(1 − 0.27)(47 − 39) + (1 − 0.27)(3.45)] = 8.923 − [5.84 + 2.5185] = **₹0.5645**; K_e = 8.7 + 0.1(10.3 − 8.7) = 8.86%; P_0 = 0.5645(1.08)/(0.0886 − 0.08) = 0.60966/0.0086 = **₹70.89**.

TYK P10 formula (ICAI answer): FCFE = Net income − [(1 − b)(capex − dep) + (1 − b)(ΔWC)], where b here = debt ratio (27%). Logic (app-written): only the equity-funded part (1 − b) of net capex and ΔWC reduces FCFE; the debt-funded part comes from new borrowing.

6.4 Dividend Discount Model versus FCFE Model

Point	DDM	FCFE
Cash flow valued	Expected **dividends** (cash paid to shareholders)	Cash **available for distribution** after capex and incremental WC
Assumption	Consistent payout, may be less than cash available	All reinvestment needs met first
Fair value	May not be fair: healthy cash with low payout → **undervaluation**	**Better measure** of value

6.5–6.5.2 Enterprise Value; EV/Sales; EV/EBITDA

→ **EV** = Market capitalisation + Long-term debt + Minority interest − Cash & cash equivalents (also minus equity investments in affiliates/other companies and long-term investments). = true economic value.

→ **Total EV** = Market cap + interest-bearing debt + minority interest − cash. **Operating EV** = Total EV − market value of non-operating assets (investments, shares in associates). **Core EV** = Operating EV − non-core assets.

→ EV considers debt and equity, **least affected by capital structure**; EV multiples more reliable than equity multiples (which see only equity claim). Denominator must represent claims of **all** claimholders.

→ **EV/Sales:** for firms with negative cash flows/earnings (cyclical), technology firms; sales least manipulable, least affected by accounting policy.

→ **EV/EBITDA:** EBITDA = proxy of cash flow to debt + equity; for **capital-intensive** firms (high D&A), acquisitions; unaffected by depreciation policy and capital structure. **Inverse** = cash return on total investment.

6.6 Valuation of Rights

Formula	Terms / logic	SM figures
$TERP = (nP_0 + S) \div (n + n_1)$	n existing shares; P_0 pre-rights price; S amount raised; n_1 new shares. Weighted average of old value and cash brought in.	TYK P2
Theoretical value of a right = Ex-Right Price − Subscription Price; Value per shareholding = (Ex-Right Price − Subscription Price) ÷ Existing Number of Shares (per right share)	What a new share is worth over its cost, spread over the old shares that carry it.	

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TYK P2: (i) 1 for 2: 5,00,000 shares at ₹4; ex-right (1,30,00,000 + 20,00,000)/15,00,000 = **₹10**; right ₹6; per share (10 − 4)/2 = **₹3**. (ii) 1 for 4: ₹8; ex-right 1,50,00,000/12,50,000 = **₹12**; right ₹4; per share (12 − 8)/4 = **₹1**. (iii) 100 shares: 150 × 10 − 200 = 1,300 = 125 × 12 − 200 = before-issue **₹1,300** → no change in wealth.

7. Valuation of Preference Shares

Formula	Terms / logic
Redeemable: Σ Dividend/$(1+r)^t$ + Maturity value/$(1+r)^n$	Fixed dividend like a bond; r = required return on preference shares.
Irredeemable: Dividend ÷ Required return	Perpetuity, like a perpetual bond.

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SM Example (7): FV ₹10,000, 10%, 3 years, r 12% → 892.86 + 797.19 + 7,829.58 = **₹9,519.63**.

8.1–8.3 Bond Basics; Bond Valuation Model; Bond Value Theorems

→ **Par value** (face, at maturity); **coupon rate** and frequency (monthly/quarterly/half-yearly/annual); **maturity period**; **redemption** — bullet (one shot) at par or premium.

Formula	Terms / logic
$V = \Sigma_{t=1..n}\ I/(1+k_d)^t + F/(1+k_d)^n = I \times PVIFA_{kd,n} + F \times PVIF_{kd,n}$	I annual interest; F principal (redemption) at maturity; n years; k_d = YTM / required return on similar bonds.

Cause	Effect
YTM = coupon	Sells at **par**
YTM > coupon	Sells at **discount**
YTM < coupon	Sells at **premium**
Longer maturity	**Greater price change** for a given change in yield

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TYK P19: 10% coupon, FV 100, redeemable ₹110 after 5 yrs: at 5% → 43.30 + 86.24 = **₹129.54**; 5.1% → **₹128.97**; 10% → 37.91 + 68.31 = **₹106.22**; 10.1% → **₹105.79**.

8.4–8.5 Yield to Maturity; Semi-annual Interest

→ **YTM** = discount rate at which PV of bond cash flows = **market price** (the bond's IRR). Find by trial and interpolation.

→ Semi-annual: $V = \Sigma_{t=1..2n}\ (1/2)/(1 + k_d/2)^t + F/(1 + k_d/2)^{2n} = 1/2 \times PVIFA_{kd/2,\ 2n} + F \times PVIF_{kd/2,\ 2n}$ — halve coupon and rate, double periods.

→ Approximate YTM as in TYK P20 answer: [11 + (100 − 97.60)/3] ÷ [(100 + 97.60)/2]; general form (app-written): [I + (F − V)/n] ÷ [(F + V)/2].

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TYK P20: (i) 11 × 2.361 + 100 × 0.693 = **₹95.27** at 13%; (ii) at 12% value ₹97.62 ≈ ₹97.60 → YTM **12%**; approx. [11 + 2.40/3]/[(100 + 97.60)/2] = **11.94%**. **TYK P21(ii):** 7.5%, ₹10,000, 2 yrs, yield 6%, half-yearly → PV interest ₹1,394 + PV principal ₹8,885 = **₹10,279**.

8.6–8.7 Price Yield Relationship; Bond Price and Time

→ Price varies **inversely** with yield (higher yield → lower PV).

→ At maturity price = par (no default). Premium bond price **falls** over time; discount bond price **rises** over time.

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TYK P21(i): current yield = coupon ÷ price = 10/110 = 9.09%; up 1% to 10.09% → price = 10/10.09% = **₹99.11**.

8.8 Duration of Bond

→ **Duration** = weighted average time to get back promised principal and YTM. Coupon bond: duration **< maturity** (TYK T1: coupons are received before maturity, so the weighted average time is shorter; ZCB duration = maturity).

→ Measures interest rate risk: higher duration = higher risk. Factors: (i) **time to maturity** — shorter → lower duration; (ii) **coupon** — higher → lower duration; (iii) **YTM** — higher → lower duration.

Formula	Terms / logic	SM figures
Macaulay Duration = $[\Sigma_{t=1..n}\ t^*c/(1+i)^t + n^*M/(1+i)^n] \div P$	Each year weighted by share of PV in price. c cash flow (coupon); i required yield; M maturity (par) value; P bond price; n time to maturity.	TYK P26, P27
Modified duration = Macaulay ÷ (1 + YTM/n)	n = compounding periods per year. = % price change for 1% change in yield ("volatility").	TYK P26: 4.247/1.17 = 3.63
Short cut (TYK P26): $D = (1+YTM)/YTM - [(1+YTM) + t(c - YTM)] \div [c((1+YTM)^t - 1) + YTM]$	YTM yield to maturity; c coupon rate; t years to maturity.	4.24 yrs
Change in price = Price × Δy × Volatility/100 (app-written, from TYK P26: 960.26 × .75 (3.63/100))	Δy in % points (75 bp = .75); price falls when yield rises.	TYK P26

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TYK P26: 16% coupon, 6 yrs, YTM 17%: price 160 × 3.589 + 1,000 × 0.390 = **₹964.24**; duration **4.247 yrs** (Σ t × PV = 4,091.20/964.40 = 4.242); volatility 4.247/1.17 = **3.63**; +75 bp → 960.26 × 0.75 × 3.63/100 = 26.142 → **₹934.118** (alt. 964.40 − 26.26 = **₹938.14**).

Note (app-written): TYK P26(d) main answer starts from ₹960.26, a figure not computed in the answer (price is ₹964.24 by formula, ₹964.40 by the table); the alternative using ₹964.40 is consistent.

8.9 Immunization

- → Rate rise → reinvestment income ↑ but bond value ↓ (and vice versa). Interest rate risk = **price risk** + **reinvestment risk**, moving in **opposite** directions.
- → Immunization: choose bonds so the two offset; portfolio value at end of holding period insensitive to rate changes. **Duration = holding period** → immunized.
- → Two bonds (TYK P27 answer): Required duration = x_1 × Duration of X + (1 − x_1) × Duration of Y, x_1 = proportion in Bond X.

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TYK P27: D_X = **1 yr**; D_Y = **3.563 yrs**; 2 = x + (1 − x)3.563 → x = **0.61** → 61% X, 39% Y; invest PV of ₹1,00,000 at 10% for 2 yrs (0.826): X ₹50,386 → **52 bonds**, Y ₹32,214 → **34 bonds**. Bond X is reinvested for one more year at the same return; modified duration may also be used.

Note (app-written): TYK P27 duration table shows PV of Bond X as ₹972.63 while the question gives price ₹972.73; the answer uses ₹972.63 for number of bonds.

8.10 Yield Curve

→ **Spot rate** = rate for a zero-coupon bond of a given maturity (time value of a single future payment). ZCB price = 10,000 ÷ (1.124) = 8,897 for year 1 (SM: 8,897 = 10,000/(1 + r_1)).

→ **Yield curve** (term structure) = YTM vs term to maturity; may slope up (long rates > short), down, flat or humped (four patterns).

→ **Forward rate** (f notation, to distinguish from spot r): f_2 = rate for the next one year after one year. Solve bond price with r_1, f_2, f_3... or from the spot curve as in SM: $f_2 = (1.1318)^2/(1.124) - 1$; $f_3 = (1.1340)^3/(1.1318)^2 - 1$.

→ Equivalent ways to value riskless cash flows: discount at the YTM, $PV[CF(t)] = CF(t)/(1 + R_t)^t$; or by the product of spot plus forward rates, $PV[CF(t)] = CF(t)/[(1 + r_1)(1 + r_2)...(1 + r_t)]$ (as printed in SM). Future spot rates are not known.

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SM Example (8.10): spot 12.40%, 13.18%, 13.40%, 13.70%, 14.10% → 2CB (₹10,000) prices 8,897; 7,807; 6,857; 5,984; 5,170. Govt securities YTM 12.400, 13.128, 13.351, 13.599, 13.901 (prices 10,000; 9,937; 10,035; 9,971; 9,948). 2-yr bond: 9,937 = 1,275/1.124 + 11,275/[1.124(1 + f_2)] → f_2 = 1,381/9,894 = **0.1396** (spot route $1.1318^2/1.124 - 1 = 0.1397$). f_3 = **0.13853** (spot 13.84%); $f_4 \approx$ **0.1461** (spot 14.60%). Forward curve: 12.40, 13.97, 13.84, 14.60, 15.71.

8.11 Term Structure Theories

→ **Expectation theory:** long-term rates are **unbiased estimators** of future short-term rates.

→ **Liquidity preference theory:** investors risk-averse; long bonds carry more interest rate risk and sacrifice liquidity → need premium → normal curve **positively sloped**.

→ **Preferred habitat (market segmentation) theory:** each investor has a preferred maturity; rates set by demand/supply in each segment; mismatch → higher rate needed to pull players out → curve may slope up, fall or be flat.

8.12 Convexity Adjustment

→ Duration assumes a **linear** price-yield relation — good only for small changes. Convexity improves the estimate.

Formula	Terms / logic
Convexity adjustment = $C^* \times (\Delta y)^2 \times 100$	Added to the duration-based % price change.
$C^* = (V_+ + V_- - 2V_0) \div [2V_0(\Delta y)^2]$	V_0 initial price; V_+ price if yield up by Δy; V_- price if yield down by Δy; Δy change in yield.

8.13–8.14 Convertible Debentures; Valuation of Warrants

→ **Conversion ratio** = equity shares per debenture; **conversion price** = price paid per share. **Conversion value** = Share price × shares per debenture.

→ **Warrant** = right to subscribe shares in a period at a stated price; issued to **sweeten** debentures. Theoretical value = (MP − E) × n (MP share price; E exercise price; n shares per warrant).

Warrant	Convertible debenture
Option **detachable** → traded separately	Not detachable
Exercised for **cash payment**	No cash payment

Measure (used in TYK answers)	Formula
Premium over conversion value	(Market price of bond − Conversion value) ÷ Conversion value
Downside risk	(Market price − Straight value) ÷ Straight value (or ÷ Market price) — fall in bond price if the stock becomes worthless
Conversion parity price	Bond price ÷ No. of shares on conversion

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TYK P22: CV 14 × 80 = 1,120; premium (1,475 − 1,120)/1,120 = **31.7%**. **TYK P23:** straight value 12 × 3.993 + 100 × 0.681 = **₹116.02**; conversion 20 × ₹4/5/6 = 80/100/120 → convert only at **₹6**. **TYK P24:** stock value 12 × 20 = **₹240**; downside (265 − 235)/235 = **12.77%** or /265 = 11.32%; conversion premium (265 − 240)/240 = **10.42%**; parity 265/20 = **₹13.25** (1.25 = 10.42% of ₹12).

8.15 Zero Coupon Bond

→ **No coupon**; issued at a **discount** to face value; one lump sum (face) at maturity = investment + accrued interest.

→ Usually **long-term** → long-range planning. Issued by banks, government, private companies; corporate ZCBs riskier (issuer strength, long maturity) but higher return. (TYK T2.)

8.16 Refunding of Bonds

Asked: Sep 2025 Q3(a) 6m

→ Issuer refunds before maturity when rates **fall**: new low-coupon bonds replace old high-coupon bonds; needs a **call feature**; cost = **call premium** (repay above face value).

→ Capital budgeting (**NPV**) decision: refund if PV of net cash savings > PV of outflows (call premium, interest in transition period etc.). New bonds usually issued before old are refunded → **overlap interest** on both.

Step	Build-up
Initial outlay	(Face + Call premium) − (New issue − Issue cost) − Tax saving on (call premium + unamortised old issue cost written off)
Annual net saving	Interest saving − tax on [interest saving + (old amortisation lost − new amortisation)]
Discount rate	After-tax cost of new debt (TYK P25 uses PVIFA 7% = 10% × (1 − 0.30)) (app-written explanation)
NPV	PV of annual net saving − Initial outlay; positive → refund

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TYK P25: cost of calling 300 + 12 = 312; net new proceeds 300 − 6 = 294; tax saving 0.30(12 + 9) = 6.3 → initial outlay **₹11.7 m**. Saving 300 × 2% = 6.00 less tax 0.30 × [6 + (9 − 6)/6] = 1.95 → **₹4.05 m** p.a.; × PVIFA(7%, 6) 4.766 = 19.30 → NPV **₹7.60 m** → refund.

8.17 Money Market Instruments

→ Short-term finance for industry, trade, government; outlet for banks' surplus funds.

→ Features: (a) **short duration**; (b) **large volume**; (c) **de-regulated** interest rates; (d) **highly liquid**; (e) **safe** (issuer strength).

→ Traditional: call and notice money, T-bills, commercial bills; new instruments give a wider choice to park funds even for a day.

ICAI ·

TYK P29: ₹20 lakh at 9%, expenses ₹50,000. (i) 5% pre-tax (₹1,00,000): 20,00,000 × 9% × P/12 − 50,000 = 1,00,000 → **P = 10 months**; (ii) break-even: = 0 → **3.33 months**.

8.17.1–8.17.2 Call/Notice Money; Treasury Bills

→ **Call money:** inter-bank, scheduled commercial banks lend/borrow **overnight (1 day)**; **notice money:** 2 to 14 days — for day-to-day surpluses/deficits.

→ **T-Bills:** short-term promissory notes of GoI issued at a **discount**; interest = discount; assured yield, negligible default risk. At present RBI issues **91, 182, 364 days** (earlier 14, 28, 91, 364 days on auction).

Formula	Terms / logic	SM figures
$Y = (F - P)/P \times 365/M \times 100$	F face value; P issue/purchase price; M actual days to maturity. Discount earned on money actually paid, annualised.	TYK P28

ICAI ·

TYK P28: 91-day, ₹100, yield 6%: $(100 - x)/x \times 365/91 \times 100 = 6 \rightarrow 0.01496x = 100 - x \rightarrow x = 100/1.01496 =$ **₹98.53**.

8.17.3–8.17.5 Commercial Bills; Certificate of Deposit; Commercial Paper

→ **Commercial bill:** arises from a genuine **credit trade** transaction; seller draws on buyer, buyer accepts; **self-liquidating**, negotiable; 3–6 months. Advantages: self-liquidating, recourse to two parties, exact date known, transparency.

→ **CD:** negotiable term deposit of commercial banks from bulk depositors at market rates; Demat or usance promissory note; front-ended, issued at discount, face value at maturity.

→ **CP:** unsecured negotiable promissory note of **high-rated corporates** for working capital, directly from market; **7 days to 1 year**; at discount; yield market-determined; introduced Jan 1990 (Vaghul Committee); by-passes banks (securitisation); partly replaces bank WC limits — no net increase in borrowing.

Formula	Terms / logic	SM figures
Discount = 100 × 15/100 × 2/12 (SM working; general form app-written: FV × rate × months/12)	Discount charged up front on face value.	Bill ₹100, 15%, 2 m → ₹2.50, fetch ₹97.50; CD ₹100, 20%, 6 m → ₹10, issue ₹90
Effective rate / yield = $(FV - SV)/SV \times$ (Days or months in a year)/M × 100	SV sale value (amount received); M period. Higher than discount rate because discount is paid up front. Same form for bill (D), CD (E) and CP (Y).	Bill **15.385%**; CD **22.22%**

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TYK P30: CP ₹10 cr, 91 days, 12.04%: 12.04% × 91/365 = 3.002% → discount 10 cr × 3.002/103.002 = ₹29,14,507 → net **₹9.70849 cr** (alt. 10 cr ÷ (1 + 12.04% × 91/365) = **₹9.70855 cr**).

8.17.6 Repo and Reverse Repo

Asked: Jan 2026 MCQ CS-1 1-3 6m

→ Raise funds by **selling securities** and agreeing to **repurchase** later at a specified price (includes interest at agreed rate). Also called **Ready Forward Contract** (sell spot/ready, buy back forward).

Point	Repo	Reverse Repo
RBI rate	Rate at which **RBI lends** to banks against G-Secs	Rate at which **banks lend** to RBI
Perspective	**Seller of securities** (acquires funds)	**Supplier of funds**; name depends on who initiated
Purpose	Meet **deficiency** of funds	Reduce **excess liquidity**
Rate level	Comparatively **higher**	Lower
Aim	Contain **inflation**	Control **money supply**

Formula (TYK P31)	Terms / logic
Dirty Price = Clean Price + Interest Accrued (= 10/100 × 100 × 262/360 in TYK P31)	Clean price excludes accrued interest; buyer pays both.
First leg (start proceed) = Nominal value × Dirty price/100 × (100 − Initial margin)/100	Cash lent against the bond, less margin (haircut).
Second leg (repayment) = Start proceed × (1 + Repo rate × days/360)	Repurchase amount = first leg + repo interest.

ICAI ·

TYK P31: 10% GoI 2028, ₹8 cr, clean 99.42, 262 days accrued → dirty 99.42 + 10 × 262/360 = **106.70**; first leg 8,00,00,000 × 106.70/100 × 98/100 = **₹8,36,52,800**; second leg × (1 + 0.0565 × 14/360) = **₹8,38,36,604**.

9.1 Role of Valuers

- → **Mergers/acquisitions/de-mergers/takeovers:** Income Tax Act, 1961 (tax payable).
- → **Slump sale/asset sale/IPR sale:** IBC, 2016 — liquidation, sale of corporate debtor's assets (fair/liquidation value).
- → **Conversion of debt/security:** RBI — inbound/outbound foreign investment.
- → **Capital reduction:** SEBI ICDR/LODR/preferential allotment — listed securities, periodically.
- → **Strategic financial restructuring:** Companies Act 2013, SARFAESI 2002, Arbitration and Conciliation Act 1996; also Ind AS and court orders.

9.2 Responsibilities of Valuers

Model Code of Conduct, Rule 12(e), Companies (Registered Valuers and Valuation) Rules, 2017 — 29 points under 8 heads:

- → **Integrity and fairness (1–4):** high integrity; honest, straightforward, forthright; true and adequate information, no misrepresentation; nothing bringing disrepute.
- → **Professional competence and due care (5–9):** due diligence, independent judgment; follow technical/professional standards; keep knowledge up to date; do not disclaim liability (except facts supplied by the company); carry out client instructions consistent with integrity, objectivity, independence.
- → **Independence and disclosure of interest (10–17):** objectivity, no bias/conflict/coercion; no assignment if he or relatives/associates not independent; independent of external influence; disclose conflicts; **no dealing in securities** of subject company once aware (SEBI PIT Regulations, 2015); no **mandate snatching** or **convenience valuations**, write to prior valuer; **no success fee**; declare past association in fairness opinions.
- → **Confidentiality (18):** no divulging confidential information without authority or legal/professional duty.
- → **Information management (19–22):** written contemporaneous records of decisions and reasons; co-operate with inspections; provide records to Registration Authority, Tribunal, Appellate Tribunal, RVO, regulators; keep working papers **3 years** (or till disposal of pending case).
- → **Gifts and hospitality (23–24):** no gifts affecting independence; no gifts/advantage to public servants to obtain work.
- → **Remuneration and costs (25–26):** transparent, reasonable fees; no undisclosed fees.
- → **Occupation, employability and restrictions (27–29):** not too many assignments; no employment unless certificate temporarily surrendered; no business inconsistent with the profession's reputation.

10. Precautions before accepting a Valuation Assignment

→ **Not a precise estimate:** many assumptions; quality ∝ time spent on data and understanding the firm.

→ **More than financial statements:** emerging businesses valued on **narrative** factors — scalability, ease of replication, growth potential, cross-sell.

→ **Middle ground** between numbers and story — focusing on only one misses opportunities.

→ **Emotions** move stock prices; ignoring this is an obstacle to getting valuation right.

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TYK T1: duration < maturity for coupon bonds — see 8.8. **TYK T2:** Zero Coupon Bonds — see 8.15.

Exam steps — dividend / CAPM valuation: K_e by CAPM (β = Cov/Var if needed) → identify D_0 vs D_1 → constant growth: $D_1/(K_e - g)$ → compare with market price → under/over-valued → buy/hold/sell.

Exam steps — two-stage / three-stage: grow D_0 year by year in high growth → terminal price at end of year $n = D_{n+1}/(K_e - g_2)$ → discount dividends and P_n at K_e (P_n by n years) → sum. For implied K_e or g: trial two rates, interpolate.

Exam steps — H-model: g_n (e.g. ROE × retention) → H = transition years ÷ 2 → $P_0 = D_0[(1 + g_n) + H(g_c - g_n)] \div (r - g_n)$ (app-written: SM formula with D_0 taken common) → % over/under-valuation vs market price.

Exam steps — FCFF / FCFE: FCFF = EBIT(1 − t) + Dep − Capex − ΔNon-cash WC → discount at WACC (target weights) → firm value − debt = equity → ÷ shares vs price. FCFE = NI + Dep − Capex − ΔWC + net borrowing (or Net income − [(1 − b)(capex − dep) + (1 − b)(ΔWC)], b = debt ratio) → discount at K_e; multistage like DDM.

Exam steps — bond value / YTM / duration: V = I × PVIFA + F × PVIF (halve rate, double periods if half-yearly) → YTM by trial + interpolation (or approx. formula) → duration = Σ t × PV of CF ÷ P_0 → Volatility = Duration/(1 + yield) → change in price = price × Δy × Volatility/100 → immunize: duration = holding period, solve weights.

Exam steps — bond refunding: initial outlay = face + call premium − net new proceeds − tax on (call premium + unamortised old cost) → annual saving = interest saving − tax on [interest saving + old amortisation − new amortisation] → discount at after-tax cost of new debt → NPV > 0 → refund.

Exam steps — money market / repo: T-bill/CP/CD yield = (FV − P)/P × 365/days → price = FV ÷ (1 + interest rate × days/365) (TYK P30 alt.). Repo: borrower = repo, lender = reverse repo → dirty = clean + coupon × days/360 → first leg = nominal × dirty/100 × (1 − margin) → second leg = first leg × (1 + repo rate × days/360).

⑤ Security Valuation — what ICAI asked (Sep 2025, Jan 2026, May 2026)

34 marks *set across the last three attempts (9.6% of marks set). Question labels and marks are ICAI's; the one-line concept descriptions are app-written.*

Attempt	Question	Marks	Concept tested
Sep 2025	Q2(a)	6	CAPM required return plus constant-growth dividend model (existing vs revised inputs): intrinsic price and buy/hold/sell advice (Numerical)
Sep 2025	Q3(a)	6	Bond refunding decision: initial outlay (call premium, issue cost, tax) and NPV of refunding 12% debentures with 10% debentures (Numerical)
Sep 2025	Q6(a)	7	FCFF valuation with WACC from a target capital structure and value per share vs market price; FCFE needed to give the same equity value (constant growth) (Numerical)
Jan 2026	MCQ CS-1 1-3	6	Repo/reverse repo of a GOI bond: classify the two parties, accrued interest and dirty price, first-leg and second-leg proceeds (MCQ)
May 2026	Q1(b)	4	H-model intrinsic value (long-run growth = terminal ROE x retention; H = half of transition period) and % overvaluation against market price (Numerical)
May 2026	Q5(a)	5	Two-stage FCFE valuation: PV of high-growth FCFE, and implied terminal growth rate for a given market price (Numerical)

What follows, exactly as ICAI printed it in the AFM study material: (1) Test Your Knowledge theory questions, with the SM paragraph ICAI's answer points to; (2) every SM illustration with ICAI's solution, grouped by SM section; (3) Test Your Knowledge practical questions with ICAI's solutions.

Ⓣ Theory questions (SM Test Your Knowledge) with ICAI answers

Theory Q1

1. Why should the duration of a coupon carrying bond always be less than the time to its maturity?

ICAI answer: "Please refer paragraph 8.8" – SM paragraph 8.8 reproduced

8.8 Duration of Bond

Duration is the weighted average time within which an investor gets back the promised principal and the promised YTM. Investment coupon bearing bond always has a duration which is lesser than its maturity. Higher the coupon rate, lesser would be the duration and higher the yield-to-maturity, lower will be the duration of a bond.

It measures how quickly a bond will repay its true cost. The longer the time it takes the greater exposure the bond has to changes in the interest rate environment and hence, higher interest rate risk. Duration is also a measure of interest rate risk – higher duration implies higher interest rate risk and lower duration means lower interest rate risk. Following are some of factors that affect bond's duration:

(i) **Time to maturity:** The shorter-maturity bond would have a lower duration and less interest rate risk and vice versa.

(ii) **Coupon rate:** Coupon payment is a key factor in calculation of duration of bonds. The higher the coupon rate, the lower is the duration and vice versa.

(iii) **Yield-to-Maturity (YTM):** Higher yield-to-maturity means lower duration and hence, lower interest rate risk and vice versa.

Although there are many formulae to calculate the duration. However, following are commonly used methods:

(a) **Macaulay Duration:** This formula measures the number of years required to recover the true cost of a bond, considering the present value of all coupon and principal payments received in the future. The formula for Macaulay duration is as follows:

$$\text{Macaulay Duration} = \frac{\sum_{t=1}^{n} \frac{t*c}{(1+i)^t} + \frac{n*M}{(1+i)^n}}{P}$$

Where,

n = Time to maturity

C = Cash flows (Coupon Amount)

i = Required yield

M = Maturity (par) value

P = Bond price

(b) Modified Duration: This is a modified version of Macaulay duration which takes into account the interest rate changes because the changes in interest rates affect duration as the yield gets affected each time the interest rate varies.

The formula for modified duration is as follows:

$$\text{Modified Duration} = \left[\frac{\text{Macaulay Duration in years}}{\left(1 + \frac{\text{YTM}}{n}\right)}\right]$$

Where

n = Number of compounding periods per year

YTM = Yield to Maturity

Theory Q2

2. Write short notes on Zero Coupon Bonds.

ICAI answer: "Please refer paragraph 8.15" – SM paragraph 8.15 reproduced

8.15 Zero Coupon Bond

As name indicates these bonds do not pay any coupon during the life of the bonds. Instead, Zero Coupon Bonds (ZCBs) are issued at discounted price to their face value, which is the amount a bond will be worth when it matures or comes due. When a ZCB matures, the investor will receive one lump sum (face value) equal to the initial investment plus interest that has been accrued on the investment made. The maturity dates on ZCBs are usually long term. These maturity dates allow an investor for a long-range planning. ZCBs issued by banks, government and private sector companies. However, bonds issued by corporate sector carry a potentially higher degree of risk, depending on the financial strength of the issuer and longer maturity period, but they also provide an opportunity to achieve a higher return.

Ⓟ Practical questions (SM Test Your Knowledge) with ICAI solutions

Practical QI

1. A company has a book value per share of ₹ 137.80. Its return on equity is 15% and it follows a policy of retaining 60% of its earnings. If the Opportunity Cost of Capital is 18%, compute the price of the share today using both Dividend Growth Model and Walter's Model.

Solution — ICAI

1. The company earnings and dividend per share after a year are expected to be:

EPS = ₹ 137.8 × 0.15 = ₹ 20.67

Dividend = 0.40 × 20.67 = ₹ 8.27

The growth in dividend would be:

g = 0.6 × 0.15 = 0.09

(a) As per Dividend Growth Model

$$\text{Perpetual growth model Formula}: P_0 = \frac{\text{Dividend}}{K_e - g}$$

$$P_0 = \frac{8.27}{0.18 - 0.09}$$

$$P_0 = ₹\ 91.89$$

(b) Walter's approach showing relationship between dividend and share price can be expressed by the following formula

$$V_c = \frac{D + \frac{R_a}{R_c}(E - D)}{R_c}$$

Where,

V_c = Market Price of the ordinary share of the company.

R_a = Return on internal retention i.e. the rate company earns on retained profits.

R_c = Capitalisation rate i.e. the rate expected by investors by way of return from particular category of shares.

E = Earnings per share.

D = Dividend per share.

Hence,

$$V_c = \frac{8.27 + \frac{0.15}{0.18}(20.67 - 8.27)}{0.18} = \frac{18.60}{0.18}$$

= ₹ 103.35

Practical Q2

2. ABC Limited's shares are currently selling at ₹ 13 per share. There are 10,00,000 shares outstanding. The firm is planning to raise ₹ 20 lakhs to Finance a new project.

Required:

What are the ex-right price of shares and the value of a right, if

(i) The firm offers one right share for every two shares held.

(ii) The firm offers one right share for every four shares held.

(iii) How does the shareholders' wealth (holding 100 shares) change from (i) to (ii)? How does right issue increases shareholders' wealth?

Solution — ICAI

2. (i) Number of shares to be issued : 5,00,000

Subscription price ₹ 20,00,000 / 5,00,000 = ₹ 4

$$\text{Ex-right Price} = \frac{₹\ 1,30,00,000 + ₹\ 20,00,000}{15,00,000} = ₹\ 10$$

Value of a Right = ₹ 10 – ₹ 4 = ₹ 6

$$\text{Value of a Right Per Share Basis} = \frac{₹10 - ₹4}{2} = ₹3$$

(ii) Subscription price ₹ 20,00,000 / 2,50,000 = ₹ 8

$$\text{Ex-right Price} = \frac{₹\ 1,30,00,000 + ₹\ 20,00,000}{12,50,000} = ₹\ 12$$

Value of a Right = ₹ 12 – ₹ 8 = ₹ 4

$$\text{Value of a Right Per Share} = \frac{₹12 - ₹8}{4} = ₹1$$

(iii) Calculation of effect of right issue on wealth of Shareholder's wealth who is holding 100 shares.

(a) When firm offers one share for two shares held.

Value of Shares after right issue (150 X ₹ 10)	₹ 1,500
Less: Amount paid to acquire right shares (50X₹ 4)	₹ 200
	₹ 1,300

(b) When firm offers one share for every four shares held.

Value of Shares after right issue (125 X ₹ 12)	₹ 1,500
Less: Amount paid to acquire right shares (25X ₹ 8)	₹ 200
	₹ 1,300

(c) Wealth of Shareholders before Right Issue ₹ 1,300

Thus, there will be no change in the wealth of shareholders from (i) and (ii).

Practical Q3

3. MNP Ltd. has declared and paid annual dividend of ₹ 4 per share. It is expected to grow @ 20% for the next two years and 10% thereafter. The required rate of return of equity investors is 15%. Compute the current price at which equity shares should sell.

Note: Present Value Interest Factor (PVIF) @ 15%:

For year 1 = 0.8696;

For year 2 = 0.7561

Solution — ICAI

3. D_0 = ₹ 4

D_1 = ₹ 4 (1.20) = ₹ 4.80

D_2 = ₹ 4 $(1.20)^2$ = ₹ 5.76

D_3 = ₹ 4 $(1.20)^2$ (1.10) = ₹ 6.336

$$P = \frac{D_1}{(1+k_e)} + \frac{D_2}{(1+k_e)^2} + \frac{TV}{(1+k_e)^2}$$

$$TV = \frac{D_3}{k_e - g} = \frac{6.336}{0.15 - 0.10} = 126.72$$

$$P = \frac{4.80}{(1+0.15)} + \frac{5.76}{(1+0.15)^2} + \frac{126.72}{(1+0.15)^2}$$

= 4.80 x 0.8696 + 5.76 x 0.7561 + 126.72 x 0.7561 = 104.34

Practical Q4

4. On the basis of the following information:

Current dividend (Do)	=	₹ 2.50
Discount rate (k)	=	10.5%
Growth rate (g)	=	2%

(i) Calculate the present value of stock of ABC Ltd.

(ii) Is its stock overvalued if stock price is ₹ 35, ROE = 9% and EPS = ₹ 2.25? Show detailed calculation. Using PE Multiple Approach and Earning Growth Model.

Solution — ICAI

4. (i) Present Value of the stock of ABC Ltd. Is:-

$$Vo = \frac{2.50(1.02)}{0.105-0.02} = ₹\ 30$$

(ii) (A) Value of stock under the PE Multiple Approach

Particulars	
Actual Stock Price	₹ 35.00
Return on equity	9%
EPS	₹ 2.25
PE Multiple (1/Return on Equity) = 1/9%	11.11
Market Price per Share	₹ 25.00

Since, Actual Stock Price is higher, hence it is overvalued.

(B) Value of the Stock under the Earnings Growth Model

Particulars	
Actual Stock Price	₹ 35.00
Return on equity	9%
EPS	₹ 2.25
Growth Rate	2%
Market Price per Share [EPS ×(1+g)]/(K_e – g) = ₹ 2.25 × 1.02/0.07	₹ 32.79

Since, Actual Stock Price is higher, hence it is overvalued.

Practical QS

5. X Limited, just declared a dividend of ₹ 14.00 per share. Mr. B is planning to purchase the share of X Limited, anticipating increase in growth rate from 8% to 9%, which will continue for three years. He also expects the market price of this share to be ₹ 360.00 after three years.

You are required to determine:

(i) the maximum amount Mr. B should pay for shares, if he requires a rate of return of 13% per annum.

(ii) the maximum price Mr. B will be willing to pay for share, if he is of the opinion that the 9% growth can be maintained indefinitely and require 13% rate of return per annum.

(iii) the price of share at the end of three years, if 9% growth rate is achieved and assuming other conditions remaining same as in (ii) above.

Calculate rupee amount up to two decimal points.

	Year-1	Year-2	Year-3
FVIF @ 9%	1.090	1.188	1.295
FVIF @ 13%	1.130	1.277	1.443
PVIF @ 13%	0.885	0.783	0.693

Solution — ICAI

5. (i) Expected dividend for next 3 years.

Year 1 (D_1) ₹ 14.00 (1.09) = ₹ 15.26

Year 2 (D_2) ₹ 14.00 $(1.09)^2$ = ₹ 16.63

Year 3 (D_3) ₹ 14.00 $(1.09)^3$ = ₹ 18.13

Required rate of return = 13% (Ke)

Market price of share after 3 years = (P_3) = ₹ 360

The present value of share

$$P_0 = \frac{D_1}{(1+ke)} + \frac{D_2}{(1+ke)^2} + \frac{D_3}{(1+ke)^3} + \frac{P_3}{(1+ke)^3}$$

$$P_0 = \frac{15.26}{(1+0.13)} + \frac{16.63}{(1+0.13)^2} + \frac{18.13}{(1+0.13)^3} + \frac{360}{(1+0.13)^3}$$

P_0 = 15.26(0.885) + 16.63(0.783) +18.13(0.693)+360(0.693)

P_0 = 13.50 + 13.02 + 12.56 + 249.48

P_0 = ₹ 288.56

(ii) If growth rate 9% is achieved for indefinite period, then maximum price of share should Mr. A willing be to pay is

$$P_0 = \frac{D_1}{(ke-g)} = \frac{₹\ 15.26}{0.13\text{-}0.09} = \frac{₹15.26}{0.04} = ₹\ 381.50$$

(iii) Assuming that conditions mentioned above remain same, the price expected after 3 years will be:

$$P_3 = \frac{D_4}{k_e - g} = \frac{D_3(1.09)}{0.13 - 0.09} = \frac{18.13 \times 1.09}{0.04} = \frac{19.76}{0.04} = ₹\ 494$$

Practical Q6

6. Piyush Loonker and Associates presently pay a dividend of Re. 1.00 per share and has a share price of ₹ 20.00.

(i) If this dividend were expected to grow at a rate of 12% per annum forever, what is the firm's expected or required return on equity using a dividend-discount model approach?

(ii) Instead of this situation in part (i), suppose that the dividends were expected to grow at a rate of 20% per annum for 5 years and 10% per year thereafter. Now what is the firm's expected, or required, return on equity?

Solution — ICAI

6. (i) Firm's Expected or Required Return on Equity

(Using a dividend discount model approach)

According to Dividend discount model approach the firm's expected or required return on equity is computed as follows:

$$K_e = \frac{D_1}{P_0} + g$$

Where,

K_e = Cost of equity share capital or (Firm's expected or required return on equity share capital)

D_1 = Expected dividend at the end of year 1

P_0 = Current market price of the share.

g = Expected growth rate of dividend.

Now, $D_1 = D_0 (1 + g)$ or ₹ 1 (1 + 0.12) or ₹ 1.12, P_0 = ₹ 20 and g = 12% per annum

$$\text{Therefore, } K_e = \frac{₹\,1.12}{₹\,20} + 12\%$$

Or, K_e = ₹ 17.6%

(ii) Firm's Expected or Required Return on Equity

(If dividends were expected to grow at a rate of 20% per annum for 5 years and 10% per year thereafter)

Since in this situation if dividends are expected to grow at a super normal growth rate g_s, for n years and thereafter, at a normal, perpetual growth rate of g_n beginning in the year n + 1, then the cost of equity can be determined by using the following formula:

$$P_0 = \sum_{t=1}^{n} \frac{Div_0 (1+g_s)^t}{(1+K_e)^t} + \frac{Div_{n+1}}{K_e - g_n}$$

Where,

g_s = Rate of growth in earlier years.

g_n = Rate of constant growth in later years.

P_0 = Discounted value of dividend stream.

K_e = Firm's expected, required return on equity (cost of equity capital).

Now,

g_s = 20% for 5 years, g_n = 10%

Therefore,

$$P_0 = \sum_{t=1}^{n} \frac{D_0 (1+0.20)^t}{(1+K_e)^t} + \frac{Div_{5+1}}{K_e - 0.10} \times \frac{1}{(1+K_e)^t}$$

$$P_0 = \frac{1.20}{(1+k_e)^1} + \frac{1.44}{(1+k_e)^2} + \frac{1.73}{(1+k_e)^3} + \frac{2.07}{(1+k_e)^4} + \frac{2.49}{(1+k_e)^5} + \frac{2.49(1+0.10)}{k_e - 0.10} \times \frac{1}{(1+k_e)^5}$$

or P_0 = ₹ 1.20 (PVF_1, K_e) + ₹ 1.44 (PVF_2, K_e) + ₹ 1.73 (PVF_3, K_e) + ₹ 2.07 (PVF_4, K_e) + ₹ 2.49 (PVF_5, K_e) + $\frac{Rs.2.74\,(PVF_5, K_e)}{K_e - 0.10}$

By trial and error we are required to find out K_e

Now, assume K_e = 18% then we will have

P_0 = ₹ 1.20 (0.8475) + ₹ 1.44 (0.7182) + ₹ 1.73 (0.6086) + ₹ 2.07 (0.5158) + ₹ 2.49 (0.4371) + ₹ 2.74 (0.4371) $\times \frac{1}{0.18 - 0.10}$

= ₹ 1.017 + ₹ 1.034 + ₹ 1.053 + ₹ 1.068 + ₹ 1.09 + ₹ 14.97= ₹ 20.23

Since the present value of dividend stream is more than required it indicates that Ke is greater than 18%.

Now, assume K_e = 19% we will have

P_0 = ₹ 1.20 (0.8403) + ₹ 1.44 (0.7061) + ₹ 1.73 (0.5934) + ₹ 2.07 (0.4986) + ₹ 2.49 (0.4190) + ₹ 2.74 (0.4190) $\times \frac{1}{0.19 - 0.10}$

= ₹ 1.008 + ₹ 1.017 + ₹ 1.026 + ₹ 1.032 + ₹ 1.043 + ₹ 12.76

= ₹ 17.89

Since the market price of share (expected value of dividend stream) is ₹ 20. Therefore, the discount rate is closer to 18% than it is to 19%, we can get the exact rate by interpolation by using the following formula:

$$K_e = LR + \frac{\text{NPV at LR}}{\text{NPV at LR} - \text{NPV at HR}} \times \Delta r$$

Where,

LR = Lower Rate

NPV at LR = Present value of share at LR

NPV at HR = Present value of share at Higher Rate

Δr = Difference in rates

$$K = 18\% + \frac{(₹\ 20.23 - ₹\ 20)}{₹\ 20.23 - ₹\ 17.89} \times 1\%$$

$$= 18\% + \frac{₹\ 0.23}{₹ 2.34} \times 1\%$$

= 18% + 0.10% = 18.10%

Therefore, the firm's expected, or required, return on equity is 18.10%. At this rate the present discounted value of dividend stream is equal to the market price of the share.

Practical Q7

7. Capital structure of Sun Ltd., as at 31.3.2003 was as under:

	(₹ in lakhs)
Equity share capital (₹ 100 each)	80
8% Preference share capital	40
12% Debentures	64
Reserves	32

Sun Ltd., earns a profit of ₹ 32 lakhs annually on an average before deduction of income-tax, which works out to 35%, and interest on debentures.

Normal return on equity shares of companies similarly placed is 9.6% provided:

(a) Profit after tax covers fixed interest and fixed dividends at least 3 times.

(b) Capital gearing ratio is 0.75.

(c) Yield on share is calculated at 50% of profits distributed and at 5% on undistributed profits.

Sun Ltd., has been regularly paying equity dividend of 8%.

Compute the value per equity share of the company assuming:

(i) 1% for every one time of difference for Interest and Fixed Dividend Coverage.

(ii) 2% for every one time of difference for Capital Gearing Ratio.

Solution — ICAI

7. (a) Calculation of Profit after tax (PAT)

		₹
Profit before interest and tax (PBIT)		32,00,000
Less: Debenture interest (₹ 64,00,000 × 12/100)		7,68,000
Profit before tax (PBT)		24,32,000
Less: Tax @ 35%		8,51,200
Profit after tax (PAT)		15,80,800
Less: Preference Dividend		
(₹ 40,00,000 × 8/100)	3,20,000	
Equity Dividend (₹ 80,00,000 × 8/100)	6,40,000	9,60,000
Retained earnings (Undistributed profit)		6,20,800

Calculation of Interest and Fixed Dividend Coverage

$$= \frac{\text{PAT + Debenture interest}}{\text{Debenture interest + Preference dividend}} \text{ or } \frac{\text{PAT + Debenture Interest Net of Tax}}{\text{Debenture interest + Preference dividend}}$$

$$= \frac{15,80,800 + 7,68,000}{7,68,000 + 3,20,000} \text{ or } \frac{15,80,800 + 4,99,200}{7,68,000 + 3,20,000}$$

$$= \frac{23,48,800}{10,88,000} \text{ or } \frac{20,80,000}{10,88,000} = 2.16 \text{ times or } 1.91 \text{ times}$$

(b) Calculation of Capital Gearing Ratio

$$\text{Capital Gearing Ratio} = \frac{\text{Fixed interest bearing funds}}{\text{Equity shareholders' funds}}$$

$$= \frac{\text{Preference Share Capital + Debentures}}{\text{Equity Share Capital + Reserves}} = \frac{40,00,000 + 64,00,000}{80,00,000 + 32,00,000} = \frac{1,04,00,000}{1,12,00,000} = 0.93$$

(c) Calculation of Yield on Equity Shares:

Yield on equity shares is calculated at 50% of profits distributed and 5% on undistributed profits:

	(₹)
50% on distributed profits (₹ 6,40,000 × 50/100)	3,20,000
5% on undistributed profits (₹ 6,20,800 × 5/100)	31,040
Yield on equity shares	3,51,040

$$\text{Yield on equity shares \%} = \frac{\text{Yield on shares}}{\text{Equity share capital}} \times 100$$

$$= \frac{3,51,040}{80,00,000} \times 100 = 4.39\% \text{ or, } 4.388\%.$$

Calculation of Expected Yield on Equity shares

(a) Interest and fixed dividend coverage of Sun Ltd. is 2.16 times but the industry average is 3 times. Therefore, risk premium is added to Sun Ltd. Shares @ 1% for every 1 time of difference.

Risk Premium = 3.00 – 2.16 (1%) = 0.84 (1%) = 0.84%

(b) Capital Gearing ratio of Sun Ltd. is 0.93 but the industry average is 0.75 times. Therefore, risk premium is added to Sun Ltd. shares @ 2% for every 1 time of difference.

Risk Premium = (0.75 – 0.93) (2%) = 0.18 (2%) = 0.36%

	(%)
Normal return expected	9.60
Add: Risk premium for low interest and fixed dividend coverage	0.84
Add: Risk premium for high interest gearing ratio	0.36
	10.80

Value of Equity Share

$$= \frac{\text{Actual yield}}{\text{Expected yield}} \times \text{Paid-up value of share} = \frac{4.39}{10.80} \times 100 = ₹\ 40.65$$

Practical Q8

8. ABC Ltd. has been maintaining a growth rate of 10 percent in dividends. The company has paid dividend @ ₹ 3 per share. The rate of return on market portfolio is 12 percent and the risk free rate of return in the market has been observed as 8 percent. The Beta co-efficient of company's share is 1.5.

You are required to calculate the expected rate of return on company's shares as per CAPM model and equilibrium price per share by dividend growth model.

Solution — ICAI

8. CAPM formula for calculation of Expected Rate of Return is :

$ER = R_f + \beta (R_m - R_f)$

$= 8 + 1.5 (12 - 8)$

$= 8 + 1.5 (4)$

$= 8 + 6$

$=14\%$ or 0.14

Applying Dividend Growth Model for the calculation of per share equilibrium price:

$$ER = \frac{D_1}{P_0} + g$$

$$0.14 = \frac{3\,(1.10)}{P_0} + 0.10$$

$$0.14 - 0.10 = \frac{3.30}{P_0}$$

$$0.04\ P_0 = 3.30$$

$$P_0 = \frac{3.30}{0.04} = ₹\ 82.50$$

Per share equilibrium price will be ₹ 82.50.

Practical Q9

9. A Company pays a dividend of ₹ 2.00 per share with a growth rate of 7%. The risk free rate is 9% and the market rate of return is 13%. The Company has a beta factor of 1.50. However, due to a decision of the Finance Manager, beta is likely to increase to 1.75. Find out the present as well as the likely value of the share after the decision.

Solution — ICAI

9. In order to find out the value of a share with constant growth model, the value of K_e should be ascertained with the help of 'CAPM' model as follows:

$$K_e = R_f + \beta\,(K_m - R_f)$$

Where,

K_e = Cost of equity

R_f = Risk free rate of return

β = Portfolio Beta i.e. market sensitivity index

K_m = Expected return on market portfolio

By substituting the figures, we get

$K_e = 0.09 + 1.5\,(0.13 - 0.09) = 0.15$ or 15%

and the value of the share as per constant growth model is

$$P_0 = \frac{D_1}{(k_e - g)}$$

Where,

P_0 = Price of a share

D1 = Dividend at the end of the year 1

Ke = Cost of equity

g = growth

$$P_0 = \frac{2.00}{(k_e - g)}$$

$$P_0 = \frac{2.00}{0.15 - 0.07} = ₹\ 25.00$$

Alternatively, it can also be found as follows:

$$\frac{2.00\ (1.07)}{0.15 - 0.07} = ₹\ 26.75$$

However, if the decision of finance manager is implemented, the beta (β) factor is likely to increase to 1.75 therefore, K_e would be

$K_e = R_f + \beta\ (K_m - R_f)$

$= 0.09 + 1.75\ (0.13 - 0.09) = 0.16$ or 16%

The value of share is

$$P_0 = \frac{D_1}{(k_e - g)}$$

$$P_0 = \frac{2.00}{0.16 - 0.07} = ₹\ 22.22$$

Alternatively, it can also be found as follows:

$$\frac{2.00\ (1.07)}{0.16 - 0.07} = ₹\ 23.78$$

Practical Q10

10. Calculate the value of share from the following information:

Profit after tax of the company	₹ 290 crores
Equity capital of company	₹ 1,300 crores
Par value of share	₹ 40 each
Debt ratio of company (Debt/ Debt + Equity)	27%
Long run growth rate of the company	8%
Beta 0.1; risk free interest rate	8.7%
Market returns	10.3%
Capital expenditure per share	₹ 47
Depreciation per share	₹ 39
Change in Working capital	₹ 3.45 per share

Solution — ICAI

10. No. of Shares = $\frac{₹\ 1{,}300\ \text{crores}}{₹\ 40}$ = 32.5 Crores

$$EPS = \frac{PAT}{\text{No.of shares}}$$

$$EPS = \frac{₹\ 290\ \text{crores}}{32.5\ \text{crores}} = ₹\ 8.923$$

FCFE = Net income – [(1-b) (capex – dep) + (1-b) (ΔWC)]

FCFE = 8.923 – [(1-0.27) (47-39) + (1-0.27) (3.45)]

= 8.923 – [5.84 + 2.5185] = 0.5645

Cost of Equity = $R_f + ß\ (R_m - R_f)$

= 8.7 + 0.1 (10.3 – 8.7) = 8.86%

$$Po = \frac{FCFE(1+g)}{K_e - g} = \frac{0.5645(1.08)}{0.0886 - .08} = \frac{0.60966}{0.0086} = ₹\ 70.89$$

Practical QII

11. Shares of Voyage Ltd. are being quoted at a price-earning ratio of 8 times. The company retains ₹ 5 per share which is 45% of its Earning Per Share.

You are required to compute

(i) The cost of equity to the company if the market expects a growth rate of 15% p.a.

(ii) If the anticipated growth rate is 16% per annum, calculate the indicative market price with the same cost of capital.

(iii) If the company's cost of capital is 20% p.a. & the anticipated growth rate is 19% p.a., calculate the market price per share.

Solution — ICAI

11. (i) Cost of Capital

Retained earnings (45%)	₹ 5 per share
Dividend (55%)	₹ 6.11 per share
EPS (100%)	₹ 11.11 per share
P/E Ratio	8 times
Market price	₹ 11.11 × 8 = ₹ 88.88

Cost of equity capital

$$= \left(\frac{Div}{Price} \times 100\right) + \text{Growth \%} = \frac{₹\ 6.11}{₹\ 88.88} \times 100 + 15\% = 21.87\%$$

(ii) Market Price = $\left(\frac{\text{Dividend}}{\text{Cost of Capital(\%) - Growth Rate(\%)}}\right)$

$$= \frac{₹\ 6.11}{(21.87-16)\%} = ₹\ 104.08 \text{ per share}$$

(iii) Market Price = $\frac{₹ 6.11}{(20-19)\%}$ = ₹ 611.00 per share

Practical Q12

12. Following Financial data are available for PQR Ltd. for the financial year 2008:

	(₹ in lakh)
8% debentures	125
10% bonds (2007)	50
Equity shares (₹ 10 each)	100
Reserves and Surplus	300
Total Assets	600
Assets Turnovers ratio	1.1
Effective interest rate	8%
Effective tax rate	40%
Operating margin	10%
Dividend payout ratio	16.67%
Current market Price of Share	₹ 14
Required rate of return of investors	15%

You are required to:

(i) Draw income statement for the year

(ii) Calculate its sustainable growth rate of earnings

(iii) Calculate the fair price of the Company's share using dividend discount model, and

(iv) What is your opinion on investment in the company's share at current price?

Solution — ICAI

12. Workings:

Asset turnover ratio	= 1.1
Total Assets	= ₹ 600
Turnover ₹ 600 lakhs × 1.1	= ₹ 660 lakhs
Effective interest rate	$= \frac{\text{Interest}}{\text{Libilities}} = 8\%$
Liabilities	= ₹ 125 lakhs + 50 lakhs = 175 lakh
Interest	= ₹ 175 lakhs × 0.08 = ₹ 14 lakh
Operating Margin	= 10%
Hence operating cost	= (1 - 0.10) ₹ 660 lakhs = ₹ 594 lakh
Dividend Payout	= 16.67%
Tax rate	= 40%

(i) Income statement

	(₹ Lakhs)
Sale	660
Operating Exp	594
EBIT	66
Interest	14
EBT	52
Tax @ 40%	20.80
EAT	31.20
Dividend @ 16.67%	5.20
Retained Earnings	26.00

(ii) SGR = ROE (1-b)

$$ROE = \frac{PAT}{NW} \text{ and } NW = ₹\,100\,lakh + ₹\,300\,lakh = 400\,lakh$$

$$ROE = \frac{₹\,31.2\,lakhs}{₹\,400\,lakhs} \times 100 = 7.8\%$$

$$SGR = 0.078(1 - 0.1667) = 6.5\% \text{ or } \frac{0.078 \times 0.8333}{1 - 0.078 \times 0.8333} = 6.95\%$$

(iii) Calculation of fair price of share using dividend discount model

$$P_o = \frac{D_0(1+g)}{k_e - g}$$

$$\text{Dividends} = \frac{₹\,5.2\,lakhs}{₹\,10\,lakhs} = ₹\,0.52$$

Growth Rate = 6.5% or 6.95%

$$\text{Hence } P_o = \frac{₹\,0.52(1+0.065)}{0.15-0.065} = \frac{₹\,0.5538}{0.085} = ₹\,6.51 \text{ or } \frac{0.52(1+0.0695)}{0.15-0.0695}$$

$$= \frac{0.5561}{0.0805} = ₹\,6.91$$

(iv) Since the current market price of share is ₹ 14, the share is overvalued. Hence the investor should not invest in the company.

Practical Q13

13. M/s X Ltd. has paid a dividend of ₹ 2.50 per share on a face value of ₹ 10 in the financial year ending on 31st March, 2009. The details are as follows:

Current market price of share	₹ 60
Growth rate of earnings and dividends	10%
Beta of share	0.75

Average market return	15%
Risk free rate of return	9%

Calculate the intrinsic value of the share.

Solution — ICAI

13. Intrinsic Value $P_0 = \frac{D1}{k-g}$

Using CAPM

$k = R_f + \beta (R_m - R_f)$

R_f = Risk Free Rate

β = Beta of Security

R_m = Market Return

$= 9\% + 0.75 (15\% - 9\%) = 13.5\%$

$P = \frac{2.5 \times 1.1}{0.135 - 0.10} = \frac{2.75}{0.035} = ₹ 78.57$

Practical Q14

14. Mr. A is thinking of buying shares at ₹ 500 each having face value of ₹ 100. He is expecting a bonus at the ratio of 1: 5 during the fourth year. Annual expected dividend is 20% and the same rate is expected to be maintained on the expanded capital base. He intends to sell the shares at the end of seventh year at an expected price of ₹ 900 each. Incidental expenses for purchase and sale of shares are estimated to be 5% of the market price. He expects a minimum return of 12% per annum.

Should Mr. A buy the share? If so, what maximum price should he pay for each share? Assume no tax on dividend income and capital gain.

Solution — ICAI

14. P.V. of dividend stream and sales proceeds

Year	Divd. /Sale	PVF (12%)	PV (₹)
1	₹ 20/-	0.893	17.86
2	₹ 20/-	0.797	15.94
3	₹ 20/-	0.712	14.24
4	₹ 24/-	0.636	15.26
5	₹ 24/	0.567	13.61
6	₹ 24/	0.507	12.17
7	₹ 24/	0.452	10.85
7	₹ 1026/- (₹ 900 x 1.2 x 0.95)	0.452	463.75
			₹ 563.68
	Less : - Cost of Share (₹ 500 x 1.05)		₹ 525.00
	Net gain		₹ 38.68

Since Mr. A is gaining ₹ 38.68 per share, he should buy the share.

Maximum price Mr. A should be ready to pay is ₹ 563.68 which will include incidental expenses. So the maximum price should be ₹ 563.68 x 100/105 = ₹ 536.84

Practical QIS

15. The risk free rate of return Rf is 9 percent. The expected rate of return on the market portfolio Rm is 13 percent. The expected rate of growth for the dividend of Platinum Ltd. is 7 percent. The last dividend paid on the equity stock of firm A was ₹ 2.00. The beta of Platinum Ltd. equity stock is 1.2.

(i) What is the equilibrium price of the equity stock of Platinum Ltd.?

(ii) How would the equilibrium price change when

- The inflation premium increases by 2 percent?
- The expected growth rate increases by 3 percent?
- The beta of Platinum Ltd. equity rises to 1.3?

Solution — ICAI

15. (i) Equilibrium price of Equity using CAPM

= 9% + 1.2(13% - 9%)

= 9% + 4.8%= 13.8%

$$P=\frac{D_1}{k_e-g}=\frac{2.00(1.07)}{0.138-0.07}=\frac{2.14}{0.068}=₹\,31.47$$

(ii) New Equilibrium price of Equity using CAPM

= 9.18% + 1.3(13% - 9.18%)

= 9.18% + 4.966% = 14.146%

$$P=\frac{D_1}{k_e-g}=\frac{2.00(1.10)}{0.14146-0.10}=\frac{2.20}{0.04146}=₹\,53.06$$

Alternatively, it can also be computed as follows:

= 11% + 1.3(15% - 11%)

= 11% + 5.2%= 16.20%

$$P=\frac{D_1}{k_e-g}=\frac{2.00(1.10)}{0.162-0.10}=₹\,35.48$$

Alternatively, if all the factors are taken separately then solution will be as follows:

(i) Inflation Premium increase by 2%. This raises R_X to 15.80%. Hence, new equilibrium price will be:

$$=\frac{2.00(1.07)}{0.158-0.07}=₹\,24.32$$

(ii) Expected Growth rate increases by 3%. Hence, revised growth rate stands at 10%:

$$=\frac{2.00(1.10)}{0.138-0.10}=₹\,57.89$$

(iii) Beta rises to 1.3. Hence, revised cost of equity shall be:

= 9% + 1.3(13% - 9%)

= 9% + 5.2% = 14.2%

As a result, New Equilibrium price shall be:

$$P=\frac{D_1}{k_e-g}=\frac{2.00(1.07)}{0.142-0.07}=₹\ 29.72$$

Practical Q16

16. SAM Ltd. has just paid a dividend of ₹ 2 per share and it is expected to grow @ 6% p.a. After paying dividend, the Board declared to take up a project by retaining the next three annual dividends. It is expected that this project is of same risk as the existing projects. The results of this project will start coming from the 4th year onward from now. The dividends will then be ₹ 2.50 per share and will grow @ 7% p.a.

An investor has 1,000 shares in SAM Ltd. and wants a receipt of at least ₹ 2,000 p.a. from this investment.

Show that the market value of the share is affected by the decision of the Board. Also show as to how the investor can maintain his target receipt from the investment for first 3 years and improved income thereafter, given that the cost of capital of the firm is 8%.

Solution — ICAI

16. Value of share at present $= \frac{D_1}{k_e-g}$

$$=\frac{2(1.06)}{0.08-0.06}=₹\ 106$$

However, if the Board implement its decision, no dividend would be payable for 3 years and the dividend for year 4 would be ₹ 2.50 and growing at 7% p.a. The price of the share, in this case, now would be:

$$P_0=\frac{2.50}{0.08-0.07}\times\frac{1}{(1+0.08)^3}=₹\ 198.46$$

So, the price of the share is expected to increase from ₹ 106 to ₹ 198.45 after the announcement of the project. The investor can take up this situation as follows:

Expected market price after 3 years $= \frac{2.50}{0.08-0.07}$ = ₹ 250.00

Expected market price after 2 years $= \frac{2.50}{0.08-0.07}\times\frac{1}{(1+0.08)}$ = ₹ 231.48

Expected market price after 1 years $= \frac{2.50}{0.08-0.07}\times\frac{1}{(1+0.08)^2}$ = ₹ 214.33

In order to maintain his receipt at least ₹ 2,000 for first 3 year, he would sell

10 shares in first year @ ₹ 214.33 for	₹ 2,143.30
9 shares in second year @ ₹ 231.48 for	₹ 2,083.32
8 shares in third year @ ₹ 250 for	₹ 2,000.00

At the end of 3rd year, he would be having 973 shares valued @ ₹ 250 each i.e. ₹ 2,43,250. On these 973 shares, his dividend income for year 4 would be @ ₹ 2.50 i.e. ₹ 2,432.50.

So, if the project is taken up by the company, the investor would be able to maintain his receipt of at least ₹ 2,000 for first three years and would be getting increased income thereafter.

Practical Q17

17. XYZ Ltd. paid a dividend of ₹ 2 for the current year. The dividend is expected to grow at 40% for the next 5 years and at 15% per annum thereafter. The return on 182 days T-bills is 11% per annum and the market return is expected to be around 18% with a variance of 24%.

The co-variance of XYZ's return with that of the market is 30%. You are required to calculate the required rate of return and intrinsic value of the stock.

Solution — ICAI

17. $\beta = \frac{\text{Covariance of Market Return and Security Return}}{\text{Variance of Market Return}}$

$\beta = \frac{30\%}{24\%} = 1.25$

Expected Return = $R_f + \beta(R_m - R_f)$

= 11% + 1.25(18% - 11%) = 11% + 8.75% = 19.75%

Intrinsic Value

Year	Dividend (₹)	PVF (19.75%,n)	Present Value (₹)
1	2.80	0.835	2.34
2	3.92	0.697	2.73
3	5.49	0.582	3.19
4	7.68	0.486	3.73
5	10.76	0.406	4.37
			16.36

PV of Terminal Value = $\frac{10.76(1.15)}{0.1975 - 0.15} \times 0.406$ = ₹ 105.77

Intrinsic Value = ₹ 16.36 + ₹ 105.77 = ₹ 122.13

Practical Q18

18. Rahul Ltd. has surplus cash of ₹ 100 lakhs and wants to distribute 27% of it to the shareholders. The company decides to buy back shares. The Finance Manager of the company estimates that its share price after re-purchase is likely to be 10% above the buyback price-if the buyback route is taken. The number of shares outstanding at present is 10 lakhs and the current EPS is ₹ 3.

You are required to determine:

(i) The price at which the shares can be re-purchased, if the market capitalization of the company should be ₹ 210 lakhs after buyback,

(ii) The number of shares that can be re-purchased, and

(iii) The impact of share re-purchase on the EPS, assuming that net *income is the same.*

Solution — ICAI

18. (i) Let P be the buyback price decided by Rahul Ltd.

Market Capitalisation after Buyback

1.1P (Original Shares – Shares Bought Back)

$$= 1.1P\left(10 \text{ lakhs} - \frac{27\% \text{ of } 100 \text{ lakhs}}{P}\right)$$

= 11 lakhs × P – 27 lakhs × 1.1 = 11 lakhs P – 29.7 lakhs

Again, 11 lakhs P – 29.7 lakhs

or 11 lakhs P = 210 lakhs + 29.7 lakhs

$$\text{or } P = \frac{239.7}{11} = ₹ 21.79 \text{ per share}$$

(ii) Number of Shares to be Bought Back :-

$$\frac{₹ 27 \text{ lakhs}}{₹ 21.79} = 1.24 \text{ lakhs (Approx.) or } 123910 \text{ share}$$

(iii) New Equity Shares:-

10 lakhs – 1.24 lakhs = 8.76 lakhs or 1000000 – 123910 = 876090 shares

$$\therefore \text{EPS} = \frac{3 \times 10 \text{ lakhs}}{8.76 \text{ lakhs}} = ₹ 3.43$$

Thus, EPS of Rahul Ltd., increases to ₹ 3.43.

Practical Q19

19. Nominal value of 10% bonds issued by a company is ₹100. The bonds are redeemable at ₹ 110 at the end of year 5. Determine the value of the bond if required yield is (i) 5%, (ii) 5.1%, (iii) 10% and (iv) 10.1%.

Solution — ICAI

19. Case (i) Required yield rate = 5%

Year	Cash Flow ₹	DF (5%)	Present Value ₹
1-5	10	4.330	43.30
5	110	0.784	86.24
Value of bond			129.54

Case (ii) Required yield rate = 5.1%

Year	Cash Flow ₹	DF (5.1%)	Present Value ₹
1-5	10	4.317	43.17
5	110	0.780	85.80
Value of bond			128.97

Case (iii) Required yield rate = 10%

Year	Cash Flow ₹	DF (10%)	Present Value ₹
1-5	10	3.791	37.91
5	110	0.621	68.31
Value of bond			106.22

Case (iv) Required yield rate = 10.1%

Year	Cash Flow ₹	DF (10.1%)	Present Value ₹
1-5	10	3.781	37.81
5	110	0.618	67.98
Value of bond			105.79

Practical Q20

20. An investor is considering the purchase of the following Bond:

Face value	₹ 100
Coupon rate	11%
Maturity	3 years

(i) If he wants a yield of 13% what is the maximum price, he should be ready to pay for?

(ii) If the Bond is selling for ₹ 97.60, what would be his yield?

Solution — ICAI

20. (i) Calculation of Maximum price

B_0 = ₹ 11 × PVIFA (13%,3) + ₹ 100 × PVIF (13%,3)

= ₹ 11 × 2.361 + ₹ 100 × 0.693 = ₹ 25.97 + ₹ 69.30 = ₹ 95.27

(ii) Calculation of yield

At 12% the value = ₹ 11 × PVIFA (12%,3) + 100 × PVIF (12%,3)

= ₹ 11×2.402 + ₹ 100×0.712 = ₹ 26.42 + ₹ 71.20 = ₹ 97.62

It the bond is selling at ₹ 97.60 which is more than the fair value, the YTM of the bond would be less than 13%. This value is almost equal to the amount price of ₹ 97.60. Therefore, the YTM of the bond would be 12%.

Alternatively

$$YTM = \frac{₹\ 11 + \frac{(₹\ 100 - ₹\ 97.60)}{3}}{\frac{(₹\ 100 + ₹\ 97.60)}{2}} = 0.1194 \text{ or } 11.94\% \text{ say } 12\%$$

Practical Q21

21. Calculate Market Price of:

(i) 10% Government of India security currently quoted at ₹ 110, but yield is expected to go up by 1%.

(ii) A bond with 7.5% coupon interest, Face Value ₹ 10,000 & term to maturity of 2 years, presently yielding 6%. Interest payable half yearly.

Solution — ICAI

21. (i) Current yield = (Coupon Interest / Market Price) X 100

(10/110) X 100 = 9.09%

If current yield go up by 1% i.e. 10.09 the market price would be

10.09 = 10 / Market Price X 100

Market Price = ₹ 99.11

(ii) Market Price of Bond = P.V. of Interest + P.V. of Principal

= ₹ 1,394 + ₹ 8,885 = ₹ 10,279

Practical Q22

22. A convertible bond with a face value of ₹ 1,000 is issued at ₹ 1,350 with a coupon rate of 10.5%. The conversion rate is 14 shares per bond. The current market price of bond and share is ₹ 1,475 and ₹ 80 respectively. What is the premium over conversion value?

Solution — ICAI

22. Conversion rate is 14 shares per bond. Market price of share ₹ 80

Conversion Value 14 x ₹ 80 = ₹ 1120

Market price of bond = ₹ 1475

Premium over Conversion Value (₹ 1475- ₹ 1120) = $\frac{355}{1120}$ x 100 = 31.7%

Practical Q23

23. Saranam Ltd. has issued convertible debentures with coupon rate 12%. Each debenture has an option to convert to 20 equity shares at any time until the date of maturity. Debentures will be redeemed at ₹ 100 on maturity of 5 years. An investor generally requires a rate of return of 8% p.a. on a 5-year security. As an investor when will you exercise conversion for given market prices of the equity share of (i) ₹ 4, (ii) ₹ 5 and (iii) ₹ 6.

Cumulative PV factor for 8% for 5 years : 3.993

PV factor for 8% for year 5 : 0.681

Solution — ICAI

23. If Debentures are not converted its value is as under: -

	PVF @ 8 %	₹
Interest - ₹ 12 for 5 years	3.993	47.92
Redemption - ₹ 100 in 5th year	0.681	68.10
		116.02

Value of equity shares:-

Market Price	No.	Total
₹ 4	20	₹ 80
₹ 5	20	₹ 100
₹ 6	20	₹ 120

Hence, unless the market price is ₹ 6 conversion should not be exercised.

Practical Q24

24. The data given below relates to a convertible bond:

Face value	₹ 250
Coupon rate	12%
No. of shares per bond	20
Market price of share	₹ 12
Straight value of bond	₹ 235
Market price of convertible bond	₹ 265

Calculate:

(i) Stock value of bond.

(ii) The percentage of downside risk.

(iii) The conversion premium

(iv) The conversion parity price of the stock.

Solution — ICAI

24. (i) Stock value or conversion value of bond

12 × 20 = ₹ 240

(ii) Percentage of the downside risk

$\frac{₹\ 265 - ₹\ 235}{₹\ 235} = 0.1277$ or 12.77% or $\frac{₹\ 265 - ₹\ 235}{₹\ 265} = 0.1132$ or 11.32%

This ratio gives the percentage price decline experienced by the bond if the stock becomes worthless.

(iii) Conversion Premium

$$\frac{\text{Market Price} - \text{Conversion Value}}{\text{Conversion Value}} \times 100$$

$$\frac{₹\ 265 - ₹\ 240}{₹\ 240} \times 100 = 10.42\%$$

(iv) Conversion Parity Price

$$\frac{\text{Bond Price}}{\text{No. of Shares on Conversion}}$$

$$\frac{₹\ 265}{20} = ₹\ 13.25$$

This indicates that if the price of shares rises to ₹ 13.25 from ₹ 12 the investor will neither gain nor lose on buying the bond and exercising it. Observe that ₹ 1.25 (₹ 13.25 – ₹ 12.00) is 10.42% of ₹ 12, the Conversion Premium.

Practical Q25

25. ABC Ltd. has ₹ 300 million, 12 per cent bonds outstanding with six years remaining to maturity. Since interest rates are falling, ABC Ltd. is contemplating of refunding these bonds with a ₹ 300 million issue of 6 year bonds carrying a coupon rate of 10 per cent. Issue cost of the new bond will be ₹ 6 million and the call premium is 4 per cent. ₹ 9 million being the unamortized portion of issue cost of old bonds can be written off no sooner the old bonds are called off. Marginal tax rate of ABC Ltd. is 30 per cent. You are required to analyse the bond refunding decision.

Solution — ICAI

25. (i) Calculation of initial outlay:-

		₹ (million)
a.	Face value	300
	Add:-Call premium	12
	Cost of calling old bonds	312
b.	Gross proceed of new issue	300
	Less: Issue costs	6
	Net proceeds of new issue	294
c.	Tax savings on call premium and unamortized cost 0.30 (12 + 9)	6.3

∴ Initial outlay = ₹ 312 million – ₹ 294 million – ₹ 6.3 million = ₹ 11.7 million

(ii) Calculation of net present value of refunding the bond:-

Saving in annual interest expenses	₹ (million)
[300 x (0.12 – 0.10)]	6.00
Less:- Tax saving on interest and amortization	
0.30 x [6 + (9-6)/6]	1.95
Annual net cash saving	4.05
PVIFA (7%, 6 years)	4.766
∴ Present value of net annual cash saving	₹ 19.30 million
Less:- Initial outlay	₹ 11.70 million
Net present value of refunding the bond	₹ 7.60 million

Decision: The bonds should be refunded

Practical Q26

26. The following data are available for a bond

Face value	₹ 1,000
Coupon Rate	16%
Years to Maturity	6
Redemption value	₹ 1,000
Yield to maturity	17%

What is the current market price, duration and volatility of this bond? Calculate the expected market price, if increase in required yield is by 75 basis points.

Solution — ICAI

26. (a) Calculation of Market price:

= 160 (PVIAF 17%,6) + 1,000 (PVIF 17%,6)

= 160 (3.589) + 1,000 (0.390) = 574.24 + 390 = 964.24

(b) Duration

Year	Cash flow	P.V. @ 17%		Proportion of bond value	Proportion of bond value x time (years)
1	160	0.855	136.80	0.142	0.142
2	160	0.731	116.96	0.121	0.246
3	160	0.624	99.84	0.103	0.309
4	160	0.534	85.44	0.089	0.356
5	160	0.456	72.96	0.076	0.380
6	1160	0.390	452.40	0.469	2.814
			964.40	1.000	4.247

Duration of the Bond is 4.247 years

Alternatively, it can also be calculated as follows:

Year	Cash flow	P.V. @ 17%		P.V. x Year
1	160	0.855	136.80	136.80
2	160	0.731	116.96	233.92
3	160	0.624	99.84	299.52
4	160	0.534	85.44	341.76
5	160	0.456	72.96	364.80
6	1160	0.390	452.40	2714.40
			964.40	4091.20

$$D = \frac{\sum_{t=1}^{n} t \times PV \text{ of } CF_t}{P_0}$$

$$D = \frac{4091.20}{964.40} = 4.242 \text{ years}$$

Alternatively, as per Short Cut Method

$$D = \frac{1+YTM}{YTM} - \frac{(1+YTM)+t(c-YTM)}{c\left[(1+YTM)^t-1\right]+YTM}$$

Where YTM = Yield to Maturity

c = Coupon Rate

t = Years to Maturity

$$= \frac{1.17}{0.17} - \frac{1.17+6(0.16-0.17)}{0.16\left[(1.17)^6-1\right]+0.17}$$

D = 4.24 years

(c) Volatility

$$\text{Volatility of the bonds} = \frac{\text{Duration}}{(1+\text{yields})} = \frac{4.247}{1.17} = 3.63 \text{ Or } = \frac{4.2422}{1.17} = 3.6258$$

(d) The expected market price if increase in required yield is by 75 basis points.

= ₹ 960.26 × .75 (3.63/100) = ₹ 26.142

Hence expected market price is ₹ 960.26 – ₹ 26.142 = ₹ 934.118

Hence, the market price will decrease

This portion can also be alternatively done as follows

= ₹ 964.40 × .75 (3.63/100) = ₹ 26.26

then the market price will be = ₹ 964.40 – 26.26 = ₹ 938.14

Practical Q27

27. Mr. A will need ₹ 1,00,000 after two years for which he wants to make one time necessary investment now. He has a choice of two types of bonds. Their details are as below:

	Bond X	Bond Y
Face value	₹ 1,000	₹ 1,000
Coupon	7% payable annually	8% payable annually
Years to maturity	1	4
Current price	₹ 972.73	₹ 936.52
Current yield	10%	10%

Advice Mr. A whether he should invest all his money in one type of bond or he should buy both the bonds and, if so, in which quantity? Assume that there will not be any call risk or default risk.

Solution — ICAI

27. Duration of Bond X

Year	Cash flow	P.V. @ 10%		Proportion of bond value	Proportion of bond value x time (years)
1	1070	0.909	972.63	1.000	1.000

Duration of the Bond is 1 year

Duration of Bond Y

Year	Cash flow	P.V. @ 10%		Proportion of bond value	Proportion of bond value x time (years)
1	80	0.909	72.72	0.077	0.077
2	80	0.826	66.08	0.071	0.142
3	80	0.751	60.08	0.064	0.192
4	1080	0.683	737.64	0.788	3.152
			936.52	1.000	3.563

Duration of the Bond is 3.563 years

Let x_1 be the investment in Bond X and therefore investment in Bond Y shall be $(1 - x_1)$. Since the required duration is 2 years the proportion of investment in each of these two securities shall be computed as follows:

$2 = x_1 + (1 - x_1)\ 3.563$

$x_1 = 0.61$

Accordingly, the proportion of investment shall be 61% in Bond X and 39% in Bond Y respectively.

Amount of investment

Bond X	Bond Y
PV of ₹ 1,00,000 for 2 years @ 10% x 61%	PV of ₹ 1,00,000 for 2 years @ 10% x 39%
= ₹ 1,00,000 (0.826) x 61% = ₹ 50,386	= ₹ 1,00,000 (0.826) x 39% = ₹ 32,214

No. of Bonds to be purchased = ₹ 50,386/₹ 972.63 = 51.80 i.e. approx. 52 bonds	No. of Bonds to be purchased = ₹ 32,214/₹ 936.52 = 34.40 i.e. approx. 34 bonds

Note: The investor has to keep the money invested for two years. Therefore, the investor can invest in both the bonds with the assumption that Bond X will be reinvested for another one year on same returns.

Further, in the above computation, Modified Duration can also be used instead of Duration.

Practical Q28

28. RBI sold a 91-day T-bill of face value of ₹ 100 at an yield of 6%. What was the issue price?

Solution — ICAI

28. Let the issue price be X

By the terms of the issue of the T-bills:

$$6\% = \frac{100 - x}{x} \times \frac{365}{91} \times 100$$

$$\frac{6 \times 91 \times x}{36{,}500} = (100 - x)$$

$$0.01496\ x = 100 - x$$

$$x = \frac{100}{1.01496} = ₹\ 98.53$$

Practical Q29

29. Wonderland Limited has excess cash of ₹ 20 lakhs, which it wants to invest in short term marketable securities. Expenses relating to investment will be ₹ 50,000.

The securities invested will have an annual yield of 9%.

The company seeks your advice

(i) as to the period of investment so as to earn a pre-tax income of 5%. (discuss)

(ii) the minimum period for the company to breakeven its investment expenditure overtime value of money.

Solution — ICAI

29. (i) Pre-tax Income required on investment of ₹ 20,00,000

Let the period of Investment be 'P' and return required on investment ₹ 1,00,000 (₹ 20,00,000 x 5%)

Accordingly,

$$(₹\ 20{,}00{,}000 \times \frac{9}{100} \times \frac{P}{12}) - ₹\ 50{,}000 = ₹\ 1{,}00{,}000$$

P = 10 months

(ii) Break-Even its investment expenditure

$$(₹\ 20{,}00{,}000 \times \frac{9}{100} \times \frac{P}{12}) - ₹\ 50{,}000 = 0$$

P = 3.33 months

Practical Q30

30. Z Co. Ltd. issued commercial paper worth ₹10 crores as per following details:

Date of issue :	16th January, 2019
Date of maturity:	17th April, 2019
No. of days:	91
Interest rate:	12.04% p.a.

What was the net amount received by the company on issue of CP? (Charges of intermediary may be ignored)

Solution — ICAI

30. The company had issued commercial paper worth ₹10 crores

No. of days Involves = 91 days

Interest rate applicable = 12.04 % p.a.

Interest for 91 days $= 12.04\% \times \frac{91\ \text{Days}}{365\ \text{Days}} = 3.002\%$

$= \text{or } ₹\ 10 \text{ crores x } \frac{3.002}{100 + 3.002} = ₹\ 29,14,507$

= or ₹ 29.14507 Lakhs

∴ Net amount received at the time of issue:- ₹ 10.00 Crores – ₹ 0.29151 Crores = ₹ 9.70849 Crores

Alternatively, it can also be computed as follows:

$$\text{Price} = \frac{₹\ 10\ \text{Crores}}{\left(1 + 12.04\% \times \frac{91\ \text{Days}}{365\ \text{Days}}\right)} = ₹\ 9.70855\ \text{Crores}$$

Practical Q31

31. Bank A enter into a Repo for 14 days with Bank B in 10% Government of India Bonds 2028 @ 5.65% for ₹ 8 crore. Assuming that clean price (the price that does not have accrued interest) be ₹ 99.42 and initial Margin be 2% and days of accrued interest be 262 days. You are required to determine

(i) Dirty Price

(ii) Repayment at maturity. (Consider 360 days in a year)

Solution — ICAI

31. **(i)** Dirty Price

= Clean Price + Interest Accrued

$$= 99.42 + 100 \times \frac{10}{100} \times \frac{262}{360} = 106.70$$

(ii) First Leg (Start Proceed)

$$= \text{Nominal Value} \times \frac{\text{Dirty Price}}{100} \times \frac{100 - \text{Initial Margin}}{100}$$

$$= ₹8,00,00,000 \times \frac{106.70}{100} \times \frac{100 - 2}{100} = ₹8,36,52,800$$

$$\text{Second Leg (Repayment at Maturity)} = \text{Start Proceed} \times \left(1 + \text{Repo rate} \times \frac{\text{No. of days}}{360}\right)$$

$$= ₹\ 8,36,52,800 \times \left(1 + 0.0565 \times \frac{14}{360}\right) = ₹\ 8,38,36,604$$

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